

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4, BL5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A CASE STUDY

Code of Conduct:

I Eddu Suchier/192524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: E. Suchier

Annexure A

Case Study

1. An enterprise network uses Kerberos for authentication, but attackers successfully perform a replay attack to gain unauthorized access. Examine how this attack could occur and propose enhancements in the authentication process to prevent such threats.

Introduction

Kerberos is an authentication protocol used in enterprise networks to securely verify users and services. It uses a **Key Distribution Center (KDC)**, tickets, session keys, and timestamps to prevent unauthorized access. However, attackers may perform a **replay attack** if valid authentication information is captured and reused.

How the Attack Occurs

When a user authenticates, Kerberos provides a **Ticket Granting Ticket (TGT)** and service tickets. The client sends a service ticket along with an authenticator containing a timestamp. An attacker who captures this information may try to resend it to the server. If the server does not properly check the timestamp or detect previously used authenticators, it may accept the replayed request and provide unauthorized access.

Prevention Methods

The enterprise can prevent replay attacks using the following measures:

1. **Timestamp validation:** Servers should reject authentication requests with old or invalid timestamps.
2. **Clock synchronization:** All systems should maintain synchronized clocks so that timestamps can be checked accurately.
3. **Replay cache:** Servers should store recently used authenticators and reject duplicate authentication requests.
4. **Strong encryption:** Modern encryption algorithms such as AES should be used, while weak algorithms should be disabled.

5. **Limited ticket lifetime:** Shorter ticket lifetimes reduce the period in which stolen tickets can be misused.
6. **Security monitoring:** Authentication logs should be monitored for unusual or repeated ticket usage.

Conclusion

A Kerberos replay attack occurs when an attacker captures valid authentication data and attempts to reuse it. Proper **timestamp checking, replay-cache mechanisms, clock synchronization, strong encryption, and continuous monitoring** can significantly reduce this threat. Therefore, correct Kerberos configuration and secure endpoint management are essential for protecting enterprise authentication.

2. A secure communication system uses a simple hash function to verify message integrity, but attackers are able to modify messages without detection. Analyze why this approach fails and propose a solution using HMAC to improve message authentication.

Introduction

A secure communication system uses a simple hash function to verify whether messages have been modified. However, attackers are able to change messages without detection. The main reason is that a **simple hash function does not use a secret key**, so anyone, including an attacker, can calculate a new hash for a modified message.

Why the Approach Fails

Suppose the sender sends:

Message + Hash(Message)

An attacker intercepts the message, modifies it, and calculates a new hash for the modified message. The receiver calculates the hash again and finds that it matches the attacker's new hash. Therefore, the receiver cannot determine whether the message came from the legitimate sender.

Thus, a simple hash provides **integrity checking but not authentication**.

Proposed Solution: HMAC

The system should use **HMAC (Hash-based Message Authentication Code)** instead of a simple hash. HMAC combines a cryptographic hash function such as SHA-256 with a **secret key shared between the sender and receiver**.

The communication becomes:

Message + HMAC(Message, Secret Key)

When the receiver gets the message, it calculates the HMAC using the same secret key. If the calculated HMAC matches the received HMAC, the message is considered authentic and unchanged.

Advantages of HMAC

- Provides **message integrity**.
- Provides **message authentication**.
- Attackers cannot generate a valid HMAC without knowing the secret key.
- Detects unauthorized modification of messages.
- HMAC-SHA-256 provides strong security for modern applications.

Conclusion

The simple hash approach fails because the hash value can be recalculated by an attacker after modifying the message. **HMAC solves this problem by combining the message with a secret key**, making it impossible for an attacker without the key to generate a valid authentication code. Therefore, HMAC-SHA-256 is a more secure solution for message authentication and integrity.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand